NAME

mbvelocitytool-qt − Interactive water sound velocity profile editor (Qt6 version).

VERSION

Version 5.0

SYNOPSIS

mbvelocitytool-qt
[
−−input=*file* {**−I***file*}
−−format=*format_id* {**−F***format_id*}
−−begin−time=*yr/mo/da/hr/mn/sc* {**−B***yr/mo/da/hr/mn/sc*}
−−end−time=*yr/mo/da/hr/mn/sc* {**−E***yr/mo/da/hr/mn/sc*}
−−display−svp−file=*svpfile* {**−S***svpfile*}
−−edit−svp−file=*svpfile* {**−W***svpfile*}
−−angle−mode=*mode*
−−threads=*number*
−−dump
−−raypaths
−−save−edit−svp=*svpfile*
−−save−swath−svp
−−save−residuals
−−export=*image.png*
−−size=*width*x*height*
−−verbose {**−V**}
−−help {**−H**}
]

DESCRIPTION

mbvelocitytool-qt is the Qt6 version of **mbvelocitytool**(1). It is used to examine sound velocity profiles (SVPs), to create new ones, and to model the effect of changing a profile on swath bathymetry. Profiles made with it are used by **mbprocess**(1) to recalculate bathymetry from raw travel times and angles. The **mbvelocitytool**(1) manual page explains the method: the raytracing, the fit of a line to each ping, the reading of the beam residuals and the three angle modes. This page describes the program itself and how it differs.

The window holds three plots that follow the size of the window, and whose shares of it can be changed by dragging the bars between them: the profiles (depth against velocity), the beam residuals (shallow up, deep down, with error bars of one standard deviation) and the raypaths without vertical exaggeration. The program takes its appearance from the desktop, including dark mode.

One profile at a time is the editable profile, drawn in the text color with a square at each node. The others are display profiles, in colors that cycle through six in the order they are loaded, and named in the legend of the plot. When a swath file is opened, these display profiles are loaded with it:

the Levitus profile
: of the first position in the file, from **mblevitus**(1), if that program is found.

the profiles stored in the file
: in formats that record them. Each different profile is shown once, named by its number and the time of the survey ping before it (or after it, when it comes first in the file). Repeats of a profile with exactly the same nodes are left out.

files named *swathfile*_**NNN.svp**
as written by **mbsvplist**(1), unless they are only copies of a stored profile.

These profiles are replaced when another swath file is opened. Profiles opened by the user and edited profiles stay.

A profile with its nodes as given is what is saved. For the raytracing a copy of it is prepared as **mbprocess** does: a node is added at 0 m if there is none, nodes that do not increase in depth are ignored, and the profile is extended to 12000 m at its last velocity. Up to 25000 pings are read; the rest of a longer file is not used, and stored profiles after that point are not found. The window reports it when this limit is reached.

mbvelocitytool-qt is built only by CMake, with the option −DbuildQt6Tools=ON.

MB-SYSTEM AUTHORSHIP

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OPTIONS

−−input=*file* **−I***file*
Swath file to read at startup.

−−format=*format_id* **−F***format_id*
The **MBIO** format id of the swath file. Default: 0, the format is found from the file name, and the program asks for it in the window if the name does not tell.

−−begin−time=*yr/mo/da/hr/mn/sc* **−B***yr/mo/da/hr/mn/sc*
Pings before this time are ignored. Default: 1962/2/21/10/30/0.

−−end−time=*yr/mo/da/hr/mn/sc* **−E***yr/mo/da/hr/mn/sc*
Pings after this time are ignored. Default: 2062/2/21/10/30/0.

−−display−svp−file=*svpfile* **−S***svpfile*
A display profile to read at startup.

−−edit−svp−file=*svpfile* **−W***svpfile*
The editable profile to read at startup. The first residuals are calculated with it.

−−angle−mode=*mode*
The angle mode: 0 do not change the beam angles, 1 adjust them using Snell's law, 2 adjust them using the sonar geometry. Default: 1.

−−threads=*number*
The number of threads for Reprocess. Default: 0, all processors. The result does not depend on this number.

−−dump
Without a window: load the data, raytrace it through the editable profile, and print the profiles found, the number of rays that failed or stopped at the bottom of the profile, and the table of beam residuals.

−−raypaths
With **−−dump**, also print the points of the raypaths.

−−save−edit−svp=*svpfile*
Without a window: write the editable profile to this file.

−−save−swath−svp
Without a window: do what **Save Swath SVP File** does, below.

−−save−residuals
Without a window: do what **Save Residuals As Offsets** does, below.

−−export=*image.png*
Without showing a window: load the data, raytrace it, and write an image of the window.

−−size=*width***x***height*
The size of the image of **−−export**. Default: 1016x668.

−−verbose −V
Print the program version and, when given again, debugging information from **MBIO**.

−−help −H
Print a description of the program and exit.

INTERACTIVE CONTROLS

Mouse, in the profile plot

Left button on a node of the editable profile
Drag the node. It stays inside the plot and between the depths of its neighbors. The first node keeps its depth and changes only in velocity.

Double-click or middle button
Add a node to the editable profile at this place.

Right button
Opens a menu of what is under the mouse:

Activate as Editing Profile
Makes a copy of the display profile the editable profile, to be changed with the mouse and applied with **Reprocess**. The original stays as a display profile. The profile it replaces becomes a display profile too, marked (edited) if it was changed, unless it is the unchanged default or a copy of a profile already shown. If the profile has more than 50 nodes, a dialog offers all of the nodes or a simplified profile, with as few nodes as keep the velocity at every left out node within a tolerance. Default: 0.25 m/s. A simplified profile changes the result of **Reprocess** slightly; all of the nodes do not.

Hide, Remove
Take the display profile off the plot, for now or for good.

Save Profile As
Writes the display profile to a file.

Delete Node
Removes the node of the editable profile. Two nodes always remain.

Add Node Here
Adds a node to the editable profile.

Show Hidden Profiles
Brings back the hidden display profiles.

When several display profiles are close to the mouse, each has a submenu under its name.

File menu

Open Display Profile
Reads one or more display profiles.

Open Editable Profile
Reads a file as the editable profile. The program offers to save changes to the previous one.

New Editable Profile
Creates the default profile of 6 nodes at 1500 m/s.

Save Editable Profile
Writes the editable profile to a file, without changing any **mbprocess** parameters.

Open Swath Sonar Data
Reads a swath file, in the background, and then raytraces it through the editable profile, creating the default one if there is none. The format comes from the file name, or is asked for. Travel times and angles are needed; if the format has none they are estimated for 1500 m/s, with a warning, and the modeling is then only suited to exporting residuals (see **mbvelocitytool**(1)).

Open Swath Sonar Data With Format
The same, asking for the format.

Save Swath SVP File
Writes the editable profile to *swathfile*.svp and sets it and the angle mode in the **mbprocess** parameter file of the swath file. The profile needs more than two nodes.

Save Residuals As Offsets
Writes the residuals of the last **Reprocess** to *swathfile*.sbo (static beam offsets) and *swathfile*.sbao (static beam angle offsets), and sets the **mbprocess** parameter file to apply the .sbo file. The program asks first if the residuals are out of date. The profile needs more than two nodes.

Quit Exits, offering to save changes to the editable profile and warning of edited display profiles that are not saved.

View menu

Plot Scaling
A dialog that can stay open, setting the Maximum Depth of the profile plot (also of the raypath plot while no data are loaded; then the raypaths set their own depth range), the Velocity Range and Velocity Center (the plot shows the center minus half of the range to the center plus half of the range), and the Residual Range (from minus to plus this value). Opening a swath file sets the Maximum Depth and Residual Range from its depths.

Show Hidden Profiles
Brings back the hidden display profiles.

Processing menu

Reprocess
Raytraces the data through the editable profile in the background, using all processors, and draws the new residuals and raypaths. The status bar then gives the time taken and any rays that failed or stopped at the bottom of the profile, and it says when the residuals are out of date because the profile or the angle mode has changed since.

Angle Mode
Do Not Change Beam Angles, Adjust Angles Using Snell's Law (the default) or Adjust Angles Using Sonar Geometry. The mode is used by **Reprocess** and written by **Save Swath SVP File**.

EXAMPLES

To work on a swath file in the window:

```
mbvelocitytool-qt −I 0030_20250208_100905.kmall
```

To print the beam residuals with a given profile, without a window:

mbvelocitytool-qt −−dump −I 0030_20250208_100905.kmall −W cast.svp

To set a profile for **mbprocess** from a script:
mbvelocitytool-qt −I 0030_20250208_100905.kmall −W cast.svp −−save−swath−svp

To write an image of the plots:
mbvelocitytool-qt −I 0030_20250208_100905.kmall −−export=plots.png −−size=1400x900

DIFFERENCES FROM MBVELOCITYTOOL

The residuals, raypaths and output files are those of **mbvelocitytool**(1) for the same data, profile and angle mode, except that a profile too short for the data, or starting below the sonar, is prepared as **mbprocess** does, so its residuals are not misleadingly flat. The other differences: **−W** applies before the first residuals; a node is taken only within 10 pixels of the mouse, and does not jump on a click; the number of nodes and of display profiles is not limited; a ping whose beams all have the same acrosstrack distance has a flat fit at its mean depth, instead of reusing the fit of the ping before; and the temporary file of the Levitus profile is not written in the current directory.

SEE ALSO

mbsystem(1), **mbvelocitytool**(1), **mbprocess**(1), **mbset**(1), **mbsvplist**(1), **mblevitus**(1)

BUGS

The legend can lie over the profiles when zoomed in. The colors of the display profiles follow their place in the list, so removing one can change the colors of the following ones. Activating a profile does not change the plot scale.